Curriculum Vitae

SeokHyeon (Lucas) Kong

lucasshkong@gmail.com | lucasshkong.github.io

RESEARCH INTERESTS

Quantum Computer Systems and Architectures, Noisy Intermediate-Scale Quantum, Variational Quantum Algorithms, Fault-Tolerant Quantum Computers, Quantum Error Correction

EDUCATION

Sungkyunkwan University (SKKU) — **Seoul, South Korea**
College of Information and Communication Engineering — *Expected February 2026*
Bachelor of Science in Electronic and Electrical Engineering — **GPA: 3.95/4.0 (4.37/4.5)**
Bachelor of Science in Advanced Semiconductor Engineering — System Architecture Track

The Pennsylvania State University — **University Park, PA**
School of Electrical Engineering and Computer Science (EECS) — *Fall 2024*
Exchange Student in Computer Engineering — **Grade: 4.0/4.0**

PUBLICATIONS

SeokHyeon Kong, Dongwhan Kim, Kiwan Meang*, Euiseong Seo*, *"*Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy*," IEEE Computer Architecture Letters*, 2025. (In Preparation)

SeokHyeon Kong and Ha-young Oh*, "Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system," *The Journal of the Korean Institute of Information and Communication Engineering*, Oct. 2023

RESEARCH EXPERIENCE

Research Mentee — **Remote**
IonQ, Advised by Dr. Denny Dahl — *08/2025-09/2025*

- Addressed the Instant Insanity problem on NISQ devices by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods.
- Applied a Variational Quantum Algorithm to the Four Corners Map Coloring problem.
- Presented two research posters in English on Constraint Satisfaction Problems in NISQ devices at Qcenter (Quantum Information Research Support Center), Korea.

Undergraduate Research Intern — **Suwon, South Korea**
ArchLab @ SKKU, Advised by Prof. Dongmoon Min — *07/2025-08/2025*

- Built foundational expertise in Quantum / Cryogenic Computer Systems through technical presentations and academic defenses.
- Explored cryogenic computer architecture - processors, interconnects, caches, and memory - as well as quantum computer systems with a focus on superconducting qubits.

Undergraduate Researcher — **University Park, PA**
Penn State CSE, Advised by Prof. Kiwan Maeng — *08/2024-09/2025*

- Conducted in-depth GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differentially private training.
- Proposed an optimization that resolves the dominant Geometric operation overhead while preserving Gaussian noise quality.
- Evaluated the real-world applicability of discrete Gaussian noise through Differentially Private Stochastic Gradient Descent (DP-SGD) experiments across multiple models.

Undergraduate Research Intern — **Pohang, South Korea**
POSTECH QCQN (Quantum Computing and Quantum Networks) — *01/2025*
- Performed 397 nm fluorescence (Doppler) and 729 nm quadrupole spectroscopy on Ca^{+} ions and enhanced system stability through targeted adjustments.
- Implemented micromotion optimization via 2D scanning with DC compensation rods to enhance ion trapping stability.

HONORS & AWARDS

Quantum Hackathon - Director's Award — *2025*

President's List (Honor Society at SKKU, **2 consecutive years**) — *2024, 2025*

Dean's List (**5 consecutive semesters**) — *SP2023, FA2023, SP2024, FA2024, SP2025*

Korea-U.S. Student Exchange Scholarship in the field of high-tech industry ($9,000) — *FA2024*

Student Success Scholarship — *2023*

TEACHIING EXPERIENCE

Academic Tutor — **Suwon, South Korea**
Sungkyunkwan University (SKKU), Intro to Electromagnetism — *09/2023-12/2023*
- Taught Samsung Semiconductor blue-collar workers studying Material-Component Engineering at SKKU, with a primary focus on problems not covered in class.

International Summer School Teaching Assistant — **Seoul, South Korea**
Sungkyunkwan University (SKKU), Engineering Mathematics 2 — *07/2023-08/2023*
- Prepared and organized class materials, including exams.

TECHNICAL SKILLS

GPU Profiling (Nsight Systems, Nsight Compute), Circuit (Cadence Virtuoso, SoC, Verilog), Programming (C, Python, CUDA, MATLAB), Quantum Computing (Qiskit), Data Analysis (NumPy, Pandas, Matplotlib), Scientific Writing (LaTeX)

CERTIFICATIONS

Qiskit Global Summer School 2025 Quantum Excellence (IBM Quantum)
CS50's Introduction to Artificial Intelligence with Python (Harvard University)
CS50's Introduction to Cybersecurity (Harvard University)